



Utrecht
University

SUPPLYING THE DUTCH MEDIA INDUSTRY WITH ZERO EMISSIONS

Future transitions
towards zero
emissions
transportation within
the Dutch media
industry supplying
sector



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Executive Summary

This study explored how suppliers in the Dutch media industry, particularly those operating in Amsterdam, can transition their vehicle fleets to comply with the city's zero-emission regulations. These regulations, set to take full effect by 2030, mandate that only vehicles emitting no harmful substances will be allowed in these zones. With current technology and availability, these are mostly electric vehicles. To comply with these new regulations the suppliers must electrify their fleet in order to meet the new regulations. However, most suppliers spend limited time driving within these zones, and this transition poses large operational and financial challenges.

In response, the *Branchevereniging Facilitaire bedrijven Filmindustrie* (BFF) was established to unify and strengthen the voice of the sector. This research, conducted by GreenScreen, the BFF, and Utrecht University, focused on how the BFF members can transition in a feasible way. This study combined surveys and interviews to assess the current landscape and stakeholder complexity, and it developed several future scenarios. The following research question was analysed:

How can the suppliers of the Dutch media industry, which are part of the BFF, transition their vehicle fleets to comply with Amsterdam's zero-emission regulations in a way that is operationally and financially feasible?

To answer this question, this study combined surveys, interviews, and data analysis. To understand the complexity of this transition, interviews were conducted with some key stakeholders, including the Municipality of Amsterdam, the Ministry of Infrastructure and Water Management, the Film Fund, a Dutch producer, and two BFF members. These interviews were done to capture the diverse perspectives and different insights that are difficult to quantify but important to understand the broader playing field.

Using a survey, data were collected on the current composition and usage of the fleet from nine BFF members. Based on this data, three transition scenarios were modelled using Excel: a Business-as-Usual (BAU) scenario, a Zero-Emission (ZE) scenario, and a Partial Compliance (PC) scenario. Each scenario models the estimated carbon emissions and financial implications up to 2035. The resulting tool/Excel was designed for reuse and will be shared with GreenScreen and the BFF to support future planning and research.

Key findings

The survey showed that there is a fleet of 92 vehicles across nine BFF members, covering 1.2 million km annually, of which only approximately 6.5% is travelled within zero-emission zones (ZEZ). The fleet is fully owned by the members, with a mix of new and second-hand vehicles. 64% of the fleet is customised, with average costs around € 25,778. Charging capacity is mentioned as a major barrier, with only three companies reporting to have charging infrastructure. Other key challenges mentioned by members are the high investment costs, limited vehicle range and difficulty of using their special interiors on EVs.

These challenges were also brought up during the interviews. Upfront investment costs, operational disruptions from charging delays and limited EV options for specialised film equipment all complicate the transition. Additionally, the case-by-case nature of the applications for the exemptions, which cost 250 euros per vehicle, adds administrative and financial burdens. Interviewees also noted that they might resort to a dual fleet, since electric vehicles do not have the range that is needed for international projects.

The projection analysis looked at the three different scenarios mentioned. Both the ZE and PC scenarios have full electrification by 2035; however, the PC scenario has more years to achieve this transition. As seen from the results, the emissions generated from the travels in ZEZ are only a limited part of the total emissions. The PC scenario has a slightly more gradual emission reduction pathway than the ZE scenario. The emissions for both scenarios drop in 2028, which is based on survey data that showed that many companies plan to upgrade vehicles around that time.

For the cost analysis, the BAU scenario has the lowest costs, but it does not meet the zero-emission regulations and is, therefore, not a realistic option. PC shows similar or lower yearly costs compared to ZE, especially if EV prices decrease in the coming years. Costs in both scenarios could be reduced further if heavy trucks, which are primarily used outside of city centres, were allowed to remain conventional. This would help companies avoid the high cost of electric heavy-duty vehicles.

Based on these research results, several recommendations can support a smoother transition to ZEZs. For example, targeted regulatory support should be offered to sectors with specialised vehicle needs through a simplified exemption process. For companies facing technical constraints or limited activity within ZEZs, extended transition timelines are advisable. A mixed-fleet strategy, deploying EVs within ZEZs while continuing to use conventional vehicles outside, can provide operational flexibility. Delaying certain EV investments may also be beneficial, allowing companies to capitalise on falling prices and the expanding second-hand market. To prevent delays caused by grid congestion or long installation times, early planning of charging infrastructure is essential. Increasing supplier awareness of available subsidies and policy developments can further accelerate adoption. Co-investing in charging infrastructure could help reduce installation and electricity costs, while collaboration with energy providers ensures that charging demands are aligned with local grid capacity. Lastly, municipalities should facilitate collective negotiations for exemptions or deadline extensions, particularly for suppliers contributing minimally to ZEZ emissions.

These recommendations can help BFF members and other stakeholders navigate the road to zero-emission compliance in a way that is feasible and financially sustainable.

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1. Introduction

The media industry includes all businesses and organisations involved in creating, producing, and distributing content intended to inform and entertain the public. It comprises various sectors such as film, television, radio, advertising, and music. This study will focus specifically on the film, television, and advertising segments, which together represent a significant share of media production activities. In recent years, the industry at large has undergone substantial transformation, driven by digitalisation, evolving lifestyles, and increasing pressure to adopt more sustainable production practices.

As part of this shift toward sustainability, various projects and policy recommendations have targeted the decarbonisation of emissions from media supply fleets, primarily through the adoption of alternative transport technologies such as electric vehicles (EVs) and hydrogen-powered vehicles (Erickson et al., 2024; European Commission, 2021). According to the European Audiovisual Observatory, CO₂ emissions come from both direct and indirect sources, with a significant share of these emissions coming from fuel consumption used in the transportation of supplies and on-site generators (European Commission, 2021). In the Dutch media industry, this transition is recent and has been accelerated by emerging regulatory pressures (see Section 1.1). In 2021 alone, Dutch film, television, and advertising production emitted an estimated 72.7 tonnes of CO₂ (Keilbach et al., 2021). These findings underscore the urgent need for “green filmmaking” practices to reduce the environmental footprint of media production (European Commission, 2021; GreenScreen, 2025; Victory, 2015).

The Dutch media industry, while relatively small in comparison to its European counterparts, has a rich history that has developed since the 19th century, with major growth during the First World War. However, the Second World War and, later, the COVID-19 pandemic severely affected its economy in terms of the number of films produced and the number of cinema attendances. While 2023 saw a resurgence in the creative media industry in the Netherlands, strongly supported by domestic initiatives, including state funding from the Netherlands Film Fund, the Dutch media industry, encompassing film, television, and advertising, operates on a relatively smaller scale compared to its European counterparts (Erickson et al., 2024; GreenScreen, 2025). As a result, many artists and suppliers in the Dutch media industry tend to work within a flexible and decentralised network, rather than under large studio systems or major industry alliances.

Within this culturally significant and diverse network, logistical support plays a crucial role from the pre-production stage through to the end of production. Suppliers are therefore essential in meeting the operational demands of the media industry. Until recently, Dutch suppliers operated independently and lacked collective representation. In contrast, bigger and more established film industries, such as the British filmmaking sector, benefit from long-standing trade unions, like the Broadcasting, Entertainment, Cinematography and Theatre Union (BECTU), and professional associations that advocate for their members, including negotiating exemptions from new government regulations. Without such structures, Dutch suppliers had limited capacity to influence policy or secure tailored arrangements. Therefore, in response to these challenges, particularly the upcoming zero-emission zone requirements set to take effect in cities like Amsterdam by 2030 (see Section 1.1), the *Branchevereniging Facilitaire bedrijven Filmindustrie* (BFF) was founded. This trade association unites suppliers across the Dutch media sector, aiming to professionalise the industry and strengthen its collective voice (GreenScreen, 2025).

1.1 Zero Emission Regulations in Amsterdam, Netherlands

Currently, Dutch municipalities are working hard towards a more sustainable future by, for example, reducing emissions and improving air quality. This is especially important in Dutch cities, where emissions tend to accumulate and air quality is often poor. To tackle this problem, zero-emission zones (ZEZ) are being introduced in the larger cities of the Netherlands. As of January 1st, 2025, sixteen municipalities have implemented ZEZ in their city centres. Another twelve municipalities have plans to follow for the near future (Zero-emissie stadslogistiek, 2025b).

One municipality that has already introduced such a zone is Amsterdam. In addition to the ZEZ, the municipality also has low-emission zones in place, as shown in Figure 1 below. The current ambition is to have a completely zero-emission city centre by 2030 (Gemeente Amsterdam, 2023).

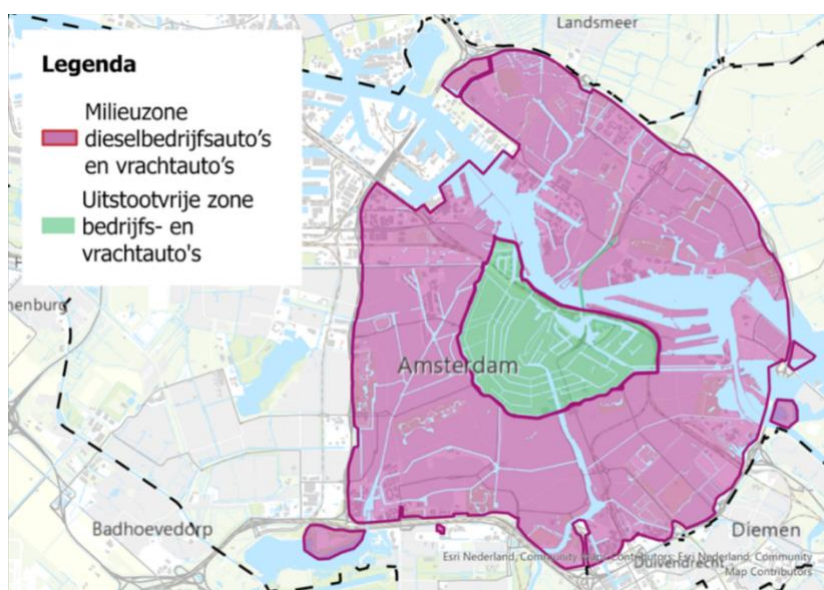


Figure 1: Low (purple)- and zero-emission zone (green) in Amsterdam (Logistiek020, 2025)

In these low-emission zones, only vehicles above a specific emission class can enter; the higher the emission class, the cleaner the vehicle. For passenger cars and vans, this currently means emission class 5 or higher, while for trucks and buses, the requirement is emission class 6. The ZEZ, however, only applies to businesses and not private vehicles, and it includes trucks and commercial vans. Within this zone in the city centre, these vehicles are not allowed to emit any harmful substances, meaning no particulate matter and CO₂ can be emitted from the exhausts of these vehicles (Gemeente Amsterdam, 2023). With current technology, only electric or hydrogen-powered vans and trucks are allowed to enter these ZEZs. A transition period has been introduced to make it possible for companies to adjust to these new regulations. Figure 2 shows what this transition period looks like. From January 1st, 2025, all trucks and vans purchased after this date must be zero-emission to enter the zone (Zero-

emissie stadslogistiek, 2025a). Different types of vehicles have different deadlines, but all timelines are aligned with the city's ambition of being fully zero-emission by January 1st, 2030.

Toegang tot zero-emissiezones					
2025	2026	2027	2028	2029	2030
Bedrijfsauto's en vrachtauto's die vanaf 1-1-2025 op kenteken worden gezet (Datum Eerste Toelating vanaf 1-1-2025) moeten een zero-emissie aandrijving hebben.*					
Bedrijfswagens (N1 en N2 voertuigen met een toegestane max. massa 3500 kg) emissieklasse 5 hebben tot 1-1-2027 toegang tot de ZE-zones.					
Bedrijfswagens (N1 en N2 voertuigen met een toegestane max. Massa 3500 kg) emissieklasse 6 en een Datum Eerste Toelating tot 1-1-2025 hebben tot 1-1-2029 toegang tot de ZE-zones.**					
Euro 6 bakwagens (N2 en N3 voertuigen met een toegestane max. massa >3500 kg) Datum Eerste Toelating (DET) van 1-1-2017 t/m 31-12-2019 hebben tot 1-1-2028 toegang tot de ZE-zones.					
Euro 6 bakwagens (N2 en N3 voertuigen met een toegestane max. massa >3500 kg) Datum Eerste Toelating (DET) van 1-1-2020 t/m 31-12-2024 hebben tot 1-1-2030 toegang tot de ZE-zones.					
Euro 6 opleggers (N2 en N3 voertuigen met een toegestane max. massa >3500 kg) Datum Eerste Toelating (DET) van 1-1-2017 t/m 31-12-2024 hebben tot 1-1-2030 toegang tot de ZE-zones.					

*Dit geldt niet voor vrijgestelde voertuigen en voertuigen die voor € 500,- of meer zijn aangepast vanwege een handicap. Deze regeling is geldig tot 1-1-2030.
 **Voorbehoud: bedrijfswagens met emissieklasse 6 hebben tot 1-1-2028 toegang tot de zero-emissiezones. Het kabinet is voornemens deze overgangsregeling te verlengen tot 1-1-2029.

Figure 2: Transition period for vehicles in the zero-emission zones (Zero-emissie stadslogistiek, 2025a)

While this transition period is meant to ease the shift, five years is still a short amount of time. Hence, various stakeholders face complex issues (see Section 1.3) in this transition towards sustainability.

1.2 Conflicts of The Different Stakeholders in The Dutch Media Industry

While the market is interested in sustainability, suppliers face many challenges and uncertainties, such as conflicting requirements, funding, and inflexibility of regulations, in making the transition to meet the zero-emission regulations by 2030 (GreenScreen, 2025). The Dutch media industry suppliers spend limited time driving in the Amsterdam city centre, and for most of them, it is neither easy nor feasible to rapidly transition to zero-emission vehicles to comply with the municipality regulations (GreenScreen, 2025).

Another concern is that there is a push for innovation in electrically powered vehicles, from government initiatives and subsidies, rather than the possibility of the use and development of alternative fuels like hydrogen (GreenScreen, 2025). As such, the infrastructure supporting hydrogen refuelling stations in the Netherlands remains limited, with only 24 stations nationwide. Although this number is gradually increasing, the expansion lags significantly behind that of electric vehicles (EV) charging infrastructure (Waterstofguide, 2023). This is despite ongoing challenges faced by EVs, such as limited range and load capacity.

While EV subsidies are contingent on meeting specific criteria related to vehicle weight, class, and engine type, subsidies for hydrogen-powered vehicles involve significantly higher costs. Companies must often bear the additional expense of developing hydrogen refuelling infrastructure themselves (RVO, n.d.a). Moreover, the purchase price of a hydrogen-powered truck is approximately €700,000, compared to €250,000 for an electric truck and diesel trucks costing over €100,000. Additionally, hydrogen fuel is around eight times more expensive than conventional diesel fuel (Waterstofguide, 2023). That is why EVs are the primary technology

considered for the transition to ZEZ, despite the challenges they still present under the current infrastructure and market practices.

For example, the supporting infrastructure and regulations for using EVs have been insufficiently integrated in the context of insurance and charging infrastructures. To ensure insurance coverage against theft, film production supplies must be kept within the buildings. However, to ensure EV insurance coverage against fires, trucks and vehicles cannot be within the buildings (GreenScreen, 2025b). This creates a nuisance for the suppliers and makes the regulations for EVs confusing (GreenScreen & BFF, 2025c). Moreover, due to the batteries, the weight-based metrics for vehicle taxing also change for EVs compared to conventional vehicles, leading to greater operational costs for the suppliers.

1.3 The Complexity of the Transition

The transition that the Dutch media industry faces is a multifaceted challenge shaped by numerous interconnected aspects. With several stakeholders involved, ranging from governmental institutions to supplier companies, different perspectives need to be considered. This elaborate hierarchy, composed of multiple firms and institutions, makes the transition a complex issue (Gell-Mann, 2002). Analysing the current and future situations requires breaking down the different aspects and gaining a holistic understanding of the industry and individual interests. Different constraints and challenges should also be part of the process in order to provide well-informed and realistic recommendations.

Although the transition toward sustainability in the film industry involves a wide range of stakeholders, this project focuses specifically on the challenges faced by the BFF suppliers and the perspectives of a broader range of stakeholders which in this research are called the task force members, and include: the municipality of Amsterdam, the Netherlands Audiovisual Producers Alliance (NAPA), the Ministry of Infrastructure and Water Management, BFF representatives, and the Netherlands Film Fund.

1.4 Research Gap

The BFF members, alongside GreenScreen, a sustainable media consultancy company, have been actively seeking engagement with government bodies to develop a realistic and actionable roadmap for electrifying their vehicle fleets in line with Amsterdam's upcoming zero-emission regulations. Previous interviews and workshops have shown the members' willingness to transition. However, the suppliers continue to face multiple barriers that hinder compliance, including logistical, financial and infrastructural challenges. This is because an understanding of the current situation and the different pathways for a just and feasible transition have not yet been mapped out. Thus, this situation highlights a diagnostic gap analysis of the desired situation and the current state of the situation between the government's sustainability targets, the capabilities of suppliers, and the insufficiency in infrastructure provided by public institutions, underscoring the need for a more coordinated and realistic approach to the transition.

1.5 Research Objective and Questions

To address this gap, GreenScreen, in collaboration with the BFF members and Utrecht University, aims to provide realistic and practical recommendations for the transition toward compliance with the 2030 zero emission regulations in Amsterdam, by analysing the current

situation through surveys, unpacking the complexities that exists amongst the stakeholders through interviews, and developing scenarios that are both aligned with Amsterdam's zero-emission policy and grounded in the operational realities and capabilities of the Dutch media industry. This objective will be achieved by answering the following research question and sub-questions:

Research Question: How can the suppliers of the Dutch media industry, that are part of the BFF, transition their vehicle fleets to comply with Amsterdam's zero-emission regulations in a way that is operationally and financially feasible?

SQ1. What is the current composition and usage profile of the vehicle fleets operated by BFF members?

SQ2. What are the perspectives, challenges, and perceived barriers faced by the task force members that hinder the sustainable transition to zero-emission fleets?

SQ3. What transition scenarios can be identified for BFF members in terms of emissions and costs in response to zero-emission regulations?

2. Research Framework

This chapter gives an overview of the framework followed in this report. The data selection and analytical framework are based on the criteria established in The Fuel Project Report (Erickson et al., 2024), a report which aims to support the London film industry's transition to low-carbon technologies. However, while this study draws from the London report, its scope is more limited due to time constraints. Specifically, it focuses on emissions generated by equipment transportation within the Dutch media industry by following a mixed-methods strategy. Following the client's recommendations, the analysis includes only suppliers that are members of the BFF. The participating BFF suppliers are listed in Table 1, while Table 2 outlines the specific criteria that define the scope of this study.

Table 1: BFF members, location, and role.

Company	Location	About
Cam-a-lot	Amsterdam	Equipment rental (camera equipment).
Eurogrip	Utrecht	Equipment rental (grip and camera equipment).
Gripwise	Amsterdam	Equipment rental (grip equipment).
Het Raam	Amsterdam	Equipment rental (camera, grip and light equipment).
Lichtmacht	Hilversum	Equipment rental (light equipment).
Lux & Co	Amsterdam	Equipment rental (light equipment).
Locatiewerk	Amsterdam	Equipment rental (vehicles, set equipment, catering).
Maloney	Amsterdam	Equipment rental (lights and cameras).
Materiaalbus	Amsterdam	Equipment rental (set equipment).

Table 2: Criteria for defining the scope of the report.

	Included	Excluded
Transport use	Transport fleets delivering to or running on productions.	Other business transport including business travel.
Suppliers	Members of the BFF.	Other suppliers that are not members of the BFF.

The following chapters outline the methodology and results for each of the sub-questions introduced in Chapter 1. This will be followed by a chapter that presents recommendations based on the findings and discussion drawn from each sub-question, aimed at addressing the main research question of this report.

Figure 3 below summarises the key steps, elaborated later in the report, to provide a clear overview of the project.

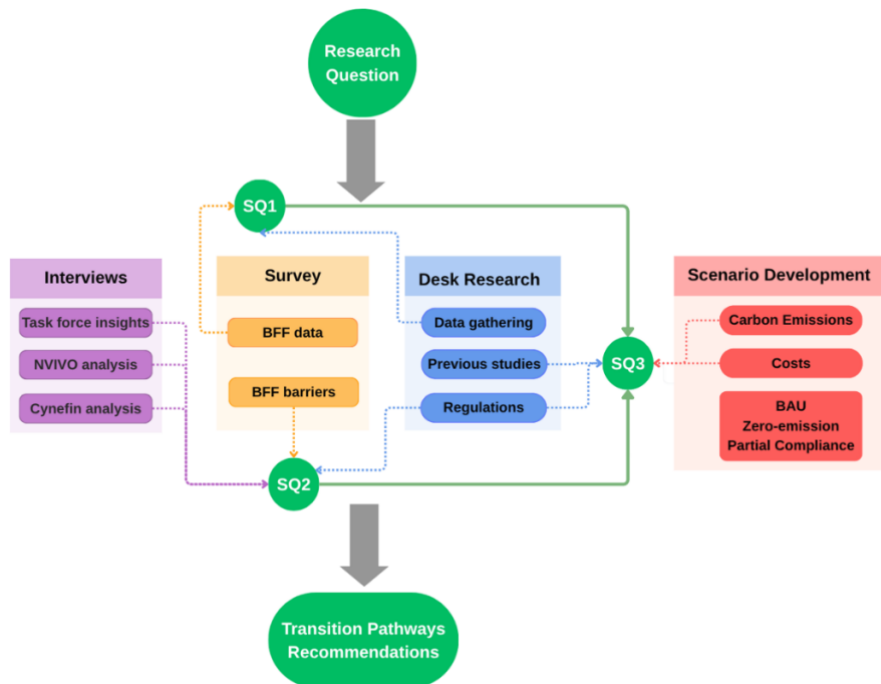


Figure 3: Research Framework

3. Sub-Question 1: Assessing the Current Fleet

3.1 Methodology

This sub-question aims to evaluate the current composition and usage of vehicle fleets operated by BFF members for transporting equipment within the Dutch media industry. To address this, a structured survey was designed to gather detailed information on each vehicle, including technical specifications, distances travelled, both in general and within ZEZ, and financial data such as purchase price, additional investments, maintenance costs, and lease contract expenses, where applicable.

The collected data will serve as input for sub-question 3. Specifically, the technical specifications and distance metrics will inform CO₂ emissions calculations for scenario development, while the financial data will support the assessment of economic feasibility in these scenarios.

Although the survey also includes a few qualitative questions to gain insight into the barriers BFF members face in transitioning toward zero-emission regulations, most qualitative findings are drawn from prior interviews conducted by GreenScreen. These interviews explored suppliers' willingness to adopt low-emission technologies and are supplemented by the task force interviews discussed in Chapter 4.

To enhance the efficiency and relevance of data collection, the research team first consulted stakeholders for feedback on the questionnaire before distributing it broadly to all BFF members, as recommended by the client. After this process, the team was informed that a similar study had previously been conducted by another member of the task force. Although the objective of that earlier study was not disclosed, the data already submitted by BFF members was shared with the research team. To ensure transparency and highlight the added value of this study, the previously provided information was pre-filled into the questionnaire. This allowed each BFF member to review, update, or complete any missing or additional data required for the current research.

The survey was designed to gather consistent and detailed information from the participating companies. It was distributed via email to 16 out of 18 supplier companies, as two had opted out at the outset. A response deadline of one week was set. Prior to distribution, the chairman of the BFF was informed to facilitate follow-up efforts and encourage participation. Companies that did not respond within the deadline were contacted again via phone and a second email. Those that remained unresponsive after follow-up were excluded from the participant list.

Given the volume of data required and its staggered reception, this step is carried out in parallel with the activities outlined in Chapter 4. The survey used to collect supplier data is included in Appendix 9.1, which provides a detailed overview of the information requested.

3.2 Results & Discussion

A total of 9 BFF member companies participated in the survey, representing a 50% share within the companies' part of the union. The companies range from small operators, e.g. companies with a two-vehicle fleet, to companies owning a twenty-six-vehicle fleet. Consequently, the provided services and operational scope of each company vary. To maintain the confidentiality of the participants, all data have been aggregated and analysed as if the fleets belong to a single, consolidated company.

Respondents reported a combined total of 92 vehicles used for transporting equipment, which will now be called the fleet. The fleet comprises various fuel and vehicle types, categorised in this chapter as light, medium and heavy trucks.

The research team conducted a visual analysis of online imagery, primarily sourced from manufacturer and logistics websites, to support the classification of vehicle types by length. This analysis considered the extra characteristics assigned to each vehicle by BFF members within their fleet records in the survey. Consequently, the length characteristic of light trucks was defined as smaller than or equal to 6.5 m, that of medium trucks as between 6.51 m and 7.6 m, and that of heavy trucks as greater than 7.61 m.

Subsequently, the share of each fuel type by Euro standard and vehicle type was calculated to provide insights into the age composition of the current fleet. Table 3 provides an overview of the fleet composition by vehicle type, whereas Table 4 shows the share of fuel type distribution across the entire fleet.

Table 3: Fleet composition by vehicle type.

Vehicle Type	Number of Vehicles	Share of Fleet
Light Trucks	37	40%
Medium Trucks	40	43%
Heavy Trucks	15	16%
Total	92	100%

Table 4: Share of Euro standard in each vehicle type and in the total fleet.

Vehicle Type	Diesel			Diesel/HVO			Electric
	Euro 4	Euro 5	Euro 6	Euro 4+	Euro 5	Euro 6	
Light Trucks	0%	22%	51%	8%	5%	5%	8%
Medium Trucks	3%	25%	53%	5%	3%	13%	0%
Heavy Trucks	0%	7%	67%	0%	0%	27%	0%
Share in the total Fleet	1%	21%	54%	5%	3%	12%	3%

The annual distance travelled varies by vehicle type. Overall, vehicles cover 1,211,495 km per year, with 6.5% of these kilometres driven within ZEZ. Table 5 shows the distances travelled per vehicle type outside and inside ZEZ.

Table 5: Annual distance travelled by vehicle type.

Vehicle Type	Annual distance (km)	Annual distance in ZEZ (km)	Share in ZEZ
Light Trucks	545,062	33,888	6.2%
Medium Trucks	422,119	37,772	8.9%
Heavy Trucks	244,313	7,440	3.0%
Total	1,211,495	79,101	6.5%

When data on kilometres travelled per vehicle were missing, the research team first estimated the values using the annual kilometres reported in the qualitative section of the questionnaire. These annual figures were then divided by the number of vehicles in the company's fleet to obtain a per-vehicle estimate. In the one case where this question was not answered, the average distance travelled per vehicle across all other companies was used as a proxy. Similarly, to estimate the distance travelled within ZEZ when this information was not provided, the research team calculated an average share of ZEZ travel relative to the total distance based on the 60 vehicles for which this information was available. This average (6.5%) was then applied to companies that did not report this information.

Moreover, to provide insights into fuel consumption by vehicle type within the current fleet composition, the average fuel consumption was calculated for each category, as shown in Table 6.

Table 6: Average fuel consumption per vehicle type and fuel type.

Vehicle Type	Diesel	Diesel/HVO
	Weighted Average FC (l/100km)	Weighted Average FC (l/100km)
Light Trucks	9	11
Medium Trucks	11	10
Heavy Trucks	16	5

In cases where fuel consumption data for a specific vehicle was missing, the research team first searched the RDW database using the vehicle's license plate. If no data were available, the fuel consumption of another vehicle of the same type was used (e.g., values from one Mercedes Sprinter were applied to another Mercedes Sprinter). When this approach was insufficient, further online searches were conducted for the specific vehicle type. If the information remained unavailable, which was the case for two vehicles, average fuel consumption values by vehicle size were sourced from grey literature. These averages were found to be 9 L/100 km for light trucks, 25 L/100 km for medium trucks, and 38 L/100 km for heavy trucks.

Results indicated that 100% of the fleet is owned by the members. Although not all respondents provided this information explicitly, ownership was assumed by the research team in cases where vehicles had customised interiors or where respondents confirmed purchasing the vehicle second-hand.

Within the fleet, 54% of vehicles were reported as newly purchased and 46% as second-hand. The absolute numbers per vehicle type are presented in Table 7. While not all members answered this question for each vehicle, missing data were supplemented based on the answers provided in the qualitative responses regarding the likelihood of purchasing vehicles new or second-hand.

Table 7: Share of new and second-hand vehicles in the fleet.

Vehicle Type	New	Second-Hand
Light Trucks	23	14
Medium Trucks	14	26
Heavy Trucks	13	2
Total	50	42

Lastly, it was reported that 64% of the vehicles in the fleet are equipped with a special interior. The absolute number of such vehicles by category, along with the average cost of these special constructions, is presented in Table 8 below.

Table 8: Vehicles with special interiors and average cost of special interiors.

Vehicle Type	No Special Interior	Special Interior	Average Cost (€)
Light Trucks	14	23	25,130
Medium Trucks	16	24	25,958
Heavy Trucks	3	12	26,246
Total	33	59	25,778

This chapter provides an overview of the current fleet composition. While the data in this chapter were presented in three broad categories to offer a more straightforward overview, sub-question three, addressed in Chapter 5, relies on detailed, vehicle-level data to enable more accurate calculations, incorporating the rest of the data provided in the survey and the assumptions outlined here.

4. Sub-Question 2: Understanding Different Perspectives

4.1 Methodology

Qualitative research is used to unpack the multilevel challenges that emerge from the complexity and chaos of transitioning to zero emissions by 2030, which is hard to quantify but relevant for an in-depth understanding of the context and limitations (Lim, 2025). These findings can be used to inform policy recommendations or good-to-best practices that are relevant for the BFF members and governmental agencies that the task force members are a part of. One way to enhance the credibility of a qualitative research design is through data triangulation, that is, employing multiple data sources or analytical perspectives to enrich and validate findings (Hammarberg et al., 2016). To apply this principle, our strategy is divided into two complementary components: *desk research* and *semi-structured interviews*.

Desk research is carried out behind a hypothetical desk, rather than in the field (Verschuren et al., 2010). The material reviewed for desk research covers the period from 2019 to 2025. 2019 covers an estimated 5-year period to observe the most recent technological and policy trends; until the current year of research, 2025, which is also when the transition phase started in the Netherlands (Jan 1st, 2025). The keyword search for literature was conducted using online search engines like Scopus, Web of Science and Google Scholar. Keywords included carbon emissions in the media industry, green filmmaking, transport decarbonisation, and electric vehicle transition in urban transport. Filtering of material was based on whether the research included empirical studies on making projections and/or transitioning from conventional vehicles to electric ones, focusing on the transportation and media sectors. This way, studies that elaborate methodologies related to the BFF's transitions are included. For grey literature, governmental and stakeholders' websites were investigated, as well as shared or recommended materials such as policy letters and the London Fuel Project methodologies and documents.

The London Fuel Project provided practice-based insights about focusing on the low-carbon transformation of the transport sector in the film and television industry (Erickson et al., 2024). The Unified Best Practices Guide, produced by Green Media Solutions LLC and Earth Angel Sustainable Production Services LLC, is the benchmark guide for sustainable production in the film and television industry (Victory, 2015). Moreover, the Creative Economy Outlook (2024), published by the UN Trade and Development Agency, discusses the issue of carbon emissions in the film industry and the path to a low-carbon transition (Creative Economy Outlook 2024, 2024). In addition, Green Film Shooting, the European Media World platform, has industry sustainability information, updated reports on energy efficiency and carbon emissions (Green Film Shooting, n.d.). The Sustainable Entertainment Alliance is a union of the world's leading film, television and streaming stakeholders dedicated to advancing sustainability initiatives (Sustainable Entertainment Alliance, n.d.), which is a platform that provides diverse industry resources and reports. The analysis, hence, incorporates international perspectives while also drawing on localised insights from the Netherlands. Relevant policy documents were collected from official websites, mainly but not limited to the following departments and organisations: Municipality of Amsterdam, Netherlands Enterprise Agency, and Ministry of Infrastructure and Water Management.

Interviews were arranged through TidyCal with the help of GreenScreen's contacts by sending a booking link via email to the interviewees, with options for English and/or Dutch to ensure a fair and inclusive procedure. The interviews were semi-structured and conducted in pairs. Interviews were conducted online through Google Meet or Zoom, taking approximately 45 minutes each. All interviews were recorded with the interviewee's permission, and transcripts

were created for the qualitative analysis. Different questions were prepared for each interview depending on the taskforce member's role and organisation, and adjusted during the interview. This way, their expertise could be harnessed. All interview questions can be found in Appendix 9.2 of the report. The interview details are provided in Table 9 below.

Table 9: Interview details.

Interviewee Reference	Role/Title	Organization	Date	Language	Duration	Format
Interviewee A	Officer	Municipality of Amsterdam	12-05-2025	English	45 minutes	Online (Google Meet)
Interviewee B	Member	BFF	13-05-2025	English	45 minutes	Online (Google Meet)
Interviewee C	Film Producer	NAPA	13-05-2025	English	45 minutes	Online (Google Meet)
Interviewee D	Member	BFF	14-05-2025	English	30 minutes	Online (Zoom)
Interviewee E	Officer	Ministry of Infrastructure and Water	16-05-2025	English	45 minutes	Online (Google Meet)
Interviewee F	Member	Netherlands Film Fund	05-06-2025	English	20 minutes	Online (Google Meet)

4.1A Qualitative Data Analysis

Qualitative data analysis (QDA) began with open coding in NVivo, as detailed below. The resulting codes were then thematically organised using the Cynefin Framework, which helps correlate key concepts and better understand a situation's complexity, such as the transition from conventional to EVs. The Cynefin Framework also serves as a decision-support tool for the policy recommendations outlined in Chapter 6. Its application supports context-sensitive strategies that are appropriate for dynamic and uncertain situations.

The Cynefin Framework, shown in Figure 4, consists of five domains, each representing a different type of situation, with corresponding characteristics and recommended types of responses (McLeod & Childs, 2013). This structure allows for a holistic categorisation of rich qualitative data, linking identified issues with suitable actions in a clear and strategic manner.

The first domain is 'Simple', where the situation is indisputable and cause-and-effect relationships are easily observable, and there is often a right answer based on best practices (Snowden & Boone, 2007). Hence, the simple domain is characterised as situations that are clear to determine the cause and effect by everyone, decisions are modelled as sensing-analysing-responding and actions are based on the best practices, which are standard and repeatable (McLeod & Childs, 2013). The second domain is 'Complicated', where nothing is self-evident. Even though solutions exist, there are multiple correct answers, and the expert has to convince the stakeholders with their knowledge (McLeod & Childs, 2013). The decision model is therefore to sense-analyse-respond and take actions of good practice that are contextual. The third domain is 'Complex', characterised by unpredictability. Causal relationships are only observed in retrospect, not in advance, as the problems of this domain are of an emergent nature (new states that are not based on any patterns). Thus, the decision model is first to experiment (probe), sense the situation, and then respond; as such, emergent strategies that come about as the situation occurs are implemented (McLeod & Childs, 2013). The fourth domain is 'Chaos', which has an absence of any link between cause and effect, creating a novel situation, meaning that right answers are dependent on acting first, then sensing and responding (McLeod & Childs, 2013). The fifth domain is 'Disorder', which is a

state of not knowing in what type of system reality fits. One's own preferences are used to assess the situation. One limitation in this complex situation is that mistakes are inevitable (Fierro et al., 2018; Nachbagauer, 2021; Snowden & Boone, 2007).

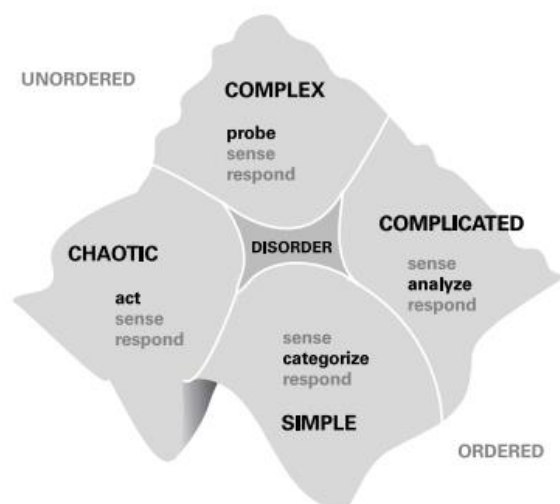


Figure 4: Cynefin Framework (Snowden & Boone, 2007)

NVivo is a QDA tool using computer-assisted software created by QSR International (Dhakal, 2022). NVivo is useful for inputting, organising, and analysing qualitative data such as interview transcripts, audio, unstructured texts, and social media posts. It allows for text analysis, like word frequencies and creating word clouds, identifying themes and discourses in data through coding, and supporting a range of research strategies, including mixed methods. The programme is accessible through Utrecht University's Solis Workspace.

The interview audio files were first converted into transcripts and cleaned, which means, spelling errors and filler words - "like, yeah", etc. - were removed, and missing information, not caught by the recording, was added. The files were then uploaded into NVivo. While (re)reading the text, the data was manually coded by themes because the questions asked differed for each interviewee. Using the information gathered from the initial open coding, insights from the initial open coding were integrated with the Cynefin framework, guiding analysis of the interview responses via the 'Probe-Sense-Analyse' model to support context-sensitive practice and policy recommendations. It should be noted that no domain is optimal over the other; they just describe the situation being faced.

The Cynefin Framework is used for different purposes in different organisations or sectors; for example, McLeod & Childs (2013) used the framework to present issues and solutions to the problem of electronic record management. However, we used the framework only as a pre-process technique, based on Snowden's (2015) four-point method. The pre-process stage is where only data about a particular issue of concern is collected through interviews (McLeod & Childs, 2013; Snowden & Boone, 2007). Additionally, the interviewees represent a minor sample of their respective organisations concerned with the sustainable transition to ZEZ; hence, they cannot be considered representative of the complete situation, as all the possible individual experiences and issues have not been completely covered. But considering that the qualitative analysis is only part of our research and the time constraints, the modifications are justified and open the way for further research.

4.2 Results & Discussion

As mentioned previously, the transition to EVs is not a smooth process. The interviews helped gain further insights into the perspectives of the stakeholders, the laws and regulations of the ZEZ, the barriers and domains where the challenges lie, and the possibilities the suppliers and taskforce members have in making the transition smoother. Table 10 summarises the information and arranges it according to the Cynefin framework. A version of the table including sub-themes to show why situations are categorised in each domain can be found in Appendix 9.3.

Table 10: Summary of the range of situations in each domain, referencing McLeod & Childs (2013).

Domain	Situations
Simple	When the situation faced is mainly covered by the national and local exemptions. The transition phase is sufficient to coordinate with national and/or local authorities to make the switch possible and feasible. <u>Exemption, subsidies, and logistic brokers procedures and costs.</u>
Complicated	Case-specific issues that require co-ordinating with a third party (logistic brokers, municipalities) to for exemptions, subsidies, or advice in a smoother electrical transition. Applying for grid connection. Building charging infrastructures and costs. Applying for an exemption is expensive, time-consuming, and difficult to implement (especially in bulk, and case specific scenarios). <u>Re-organisation of business operations due to the switch to electric vehicles.</u>
Complex	Cultural shift towards concerns over costs and politics. Further investment into research and development of electric vehicles. Lock-in with transition options of the BFF members. Cooperation with different stakeholders (grid operators and municipalities) for case-specific scenarios not covered by the exemptions.
Chaotic	Cultural shift towards concerns over costs and politics. Further investment into research and development of electric vehicles. Lock-in with transition options of the BFF members, makes the situation unpredictable and difficult to manage.

4.2A Policies

Following the first Green Deal in 2014, where multiple Dutch municipalities and the government signed an agreement on emission city logistics (Interviewee E). Followed by a climate conference in 2019, the Netherlands' goal was to have 30 to 40 ZEZ in the country by 2025 (Zero-emissie stadslogistiek, 2025c). In the next initiative for greener cities and better air quality, the experiences of the starting phase of the project were evaluated, and another climate agreement took place in 2021, where more concrete measures were established. By this time, there were over 200 different participating stakeholders, from governments, entrepreneurs, industry parties, different interest groups and knowledge institutes (Interviewee E; Zero-emissie stadslogistiek, 2025c).

To facilitate the transition, the Dutch national government has provided exemptions based on several criteria, as explained by both the municipality and the ministry's representatives. The exemptions are divided into two main categories, based on the authority responsible for granting them. The first category includes exemptions issued by the national government. These exemptions apply to Euro 5 and Euro 6 vehicles and allow a transition period until 2030. They are tied to the vehicle's license plate and are established under national law (Ontheffing Aanvragen, 2025). This means that exemptions are granted per vehicle. The second category includes case-specific exemptions granted by the municipality, having a more local role, in collaboration with the RDW. These exemptions cover situations such as financial hardship,

imminent retirement, and short-term exemptions, like one-day exemptions and up to twelve days per city per year (Ontheffing Aanvragen, 2025).

These exemptions cover the major issues companies may encounter in the attempt to transition. Every municipality in the Netherlands has the same regulations for the ZEZ, implying that the exemptions are granted based on merit. This principle does not apply in the case of an exemption for unforeseen circumstances, for example, if the company owner is a few years before retirement and does not have enough time to recover the costs incurred during the transition (Interviewee E; Zero-emissie stadslogistiek, 2025a). Such exemptions are handled by the local municipality, on top of the state-accepted exemptions. The ministry also provides corporate and local subsidies for electric trucks (Interviewee E; Netherlands Chamber of Commerce, n.d.) and logistical advice for each municipality (Zero-emissie stadslogistiek, 2025c).

According to the municipality of Amsterdam, other exemptions are due to the unavailability of electric vehicles in the market or delays in expanding the grid and building charging infrastructure. For the former, the municipality considers market forecasts to make more accurate predictions. To get this type of exemption, there should be fewer than two companies offering electric cars with certain standards at that point in time. For instance, if the forecasts show that this is the case for the next three years, the company can get the exemption for those three years. The same goes for the charging infrastructure (Zero-emissie stadslogistiek, 2025a).

In general, the municipalities and the Ministry of Infrastructure and Water work together, although the Ministry is the one that coordinates the regulations. There is a separate organisation, run by multiple municipalities, called Emission Counter (Emissie Loket), where companies can request exemptions in a more user-friendly way. However, in the end, the municipality needs to handle the final exemption (Gemeente Amsterdam, 2025). To apply for an exemption, companies pay €250 due to the time-consuming evaluation process. Calculations for granting exemptions are performed by an independent third party that collaborates with the government (Ontheffing Aanvragen, 2025). This evaluation helps companies know whether they meet the criteria or not. Proving that a company cannot afford the transition is challenging because the evaluation is performed on a case-by-case basis. This check is repeated every year to see if the situation has stayed the same or if the company can manage to invest in the transition.

4.2B Barriers and Challenges

A number of interviewees pointed out that the upfront cost of switching to EVs is a major obstacle. The cost for electric trucks (especially when customised) for the film industry is significantly higher than that of their diesel counterparts in terms of purchasing and, therefore, offered rental pricing. While diesel trucks can be rented for around €80 per day, electric trucks cost roughly €200 per day, creating financial pressure, particularly since producers tend to be reluctant to pay for these additional expenses (Interviewee A & B). Additionally, each exemption application burdens the respective company with a cost of €250, which might not be feasible for the case of a small company or a company with many vehicles. This also creates a point of conflict on who should share the burden of costs.

Apart from the expenses, multiple interviewees also noted that the transition requires companies to reorganise their entire logistical system. The limited range and time-consuming charging of electric vehicles directly disrupted the original workflow, especially for the film production vehicles, which often remain on set for extended periods, frequently not departing

until late at night, so there aren't many opportunities for charging (Interviewee B). Moreover, hotels often lack sufficient fast charging stations for fleets, so they may need to wait a long time before continuing their journey. With fast charging stations or direct current fast chargers (DCFC), the charging of an electric truck will take between 45 and 100 minutes, depending on the type. However, these stations aren't yet widely available, with slower charging stations, the approximate charging time can vary from 5 to 20 hours, depending on the type (Gexa Energy, 2025).

In the Netherlands, the problems with grid congestion and electricity shortages further complicate the situation. Some interviewees mentioned that even if they are willing to switch to EVs, the required grid connection or infrastructure cannot be secured at their base location (Interviewee E). In principle, batteries or solar panels could offer alternatives, but these are only feasible in very specific cases. Furthermore, since exemptions are granted per vehicle, companies with multiple vehicles must submit applications for each one individually, involving time and money.

Moreover, operational constraints specific to the film industry make the transition more complex. According to one BFF member: Most vehicles used on location are highly specialised, additions such as grip trucks, camera cranes, or makeup units are difficult to replace and generally not shareable. Multiple companies may work simultaneously on the same set, so sharing vehicles is unrealistic. Another challenge noted by a BFF member is the range and flexibility. While the range on some EVs is about a maximum of 350 km, this is not enough for long-distance shoots in other countries like Spain or Italy. Companies must often switch back to diesel vehicles for these journeys, requiring dual fleet operations and increasing logistical complexity and labour costs. Larger companies like IKEA benefit from having internal transition teams, so they can better access subsidies and establish contacts with regulatory authorities. In contrast, smaller companies like many companies in the BFF union lack these resources and institutional knowledge, making the transition critically more difficult (Interviewee A).

Finally, it is mentioned that the industry's focus has shifted in recent years. Five years ago, sustainability was a hot topic, but current inflation and rising wage demands have made cost management more of a priority. Green initiatives are a secondary consideration for many producers, as mentioned in multiple interviews; therefore, paying extra for EVs might not be well-received. Although the zero-emission transition is widely recognised and supported in principle, BFF members still face a set of complex and connected challenges, including financial, logistical, regulatory, and technical issues, which make implementation notably difficult without tailored support and long-term policy clarity.

4.2C Cynefin Framework Analysis

The transition period, as discussed in Chapter 1, is set because the government acknowledges the impossible nature of electrifying a whole fleet overnight. The regulations are not intended to place a financial burden on companies, but rather to achieve a better environmental impact, which can also be seen in the implementation agenda of the municipality of Amsterdam, where the main purpose of these measures is improving the air quality in the city (Gemeente Amsterdam, 2023). Given the implemented regulations, companies in the Netherlands, including companies from the BFF, are required to make the transition, which comes with high investment costs and several other challenges, further analysed in this section according to how the situation could range from simple to chaotic. The framework was used by reflecting on the situation (e.g., what was the situation, what changes had to be made, how easy it is to understand it, and how straightforward the solution is to

find). For example, the simple domain refers to there being a right, standardised answer that can be followed by the BFF members (McLeod & Childs, 2013). Similar coding of the domains is mentioned in Table 10 and expanded upon in this section.

The authorities recognise that a transition phase needs to be implemented and that not everyone is capable of meeting the requirements of this transition. Thus, a time limit of four years and exemptions are provided. Suppliers facing challenges that are primarily addressed by the exemptions fall into the simple domain, as there is a standard procedure that can be applied and agreed upon by all parties. For instance, when specific conditions are met, such as having adequate financial resources, receiving subsidies for EVs, or finding a suitable EV, the transition becomes straightforward. At most, only the business operating procedures need to be redesigned to adapt to the new situation, which may involve changing logistical plans.

The situation goes to complicated when finding a solution suited for a specific company, as expert analysis is needed, and there are multiple solutions that change on a case-by-case basis. The complication with the current range on EVs requires reorganisation of business logistics, which requires spending time and energy, strategic planning, and expertise in telematics and cost management. Applying for grid connection or investing in building more charging infrastructure is also a complicated process, as planning and co-operation with grid operators and the municipality are required.

The process of the exemptions can border on complicated and complex. For example, bulk exemptions are hard to materialise by law, indicating that such exemptions may never be implemented (Interviewee A). This has direct implications for the BFF's transition, as securing an exemption for the entire union may prove unfeasible. The more exemptions that are allowed or the more case-specific the situation, the more complex and difficult the regulations become to enforce, as stated by the interviewee from the Municipality. A balance is needed in how accessible exemptions are to make the transition happen without placing an excessive burden on companies, but this requires a long and experimental process from the involved parties to determine a good solution.

When it comes to EVs, the situation can range from simple to complex, largely shaped by their current availability, performance and developments in the field. Cost-effective solutions for the switch to EVs do exist; however, as stated above, this process requires cooperation with different stakeholders and collaboration for a whole system scenario planning, as the switch would involve changes to operations.

The situations that range from complex to chaotic mainly have to do with situations that involve multiple perspectives. For example, the cultural shift towards sustainable norms and attitudes is something determined by socio-economic trends, as seen from one interview, the current priorities are managing the costs and budget limits over sustainability (Interviewee C). Furthermore, the BFF members are locked into their options for transitioning within the situation of the charging infrastructure, the lack of feasible alternatives (like shared electric mobility) due to their workflows, and the pressure to make the transition. This situation borders between being complex to chaotic as innovative and emerging practices set the new practices, and there are no right solutions (McLeod & Childs, 2013). This makes it difficult to predict or manage the situation through the standard methods.

4.2D Possibilities

Transformation is inevitable, and how and when to undertake it is the key focus for everyone involved. Currently, the government and municipalities are actively trying to help companies

with this transition. For example, companies can consult transport brokers via their official websites for personal advice on transition plans. There are also some transition tax exemptions in place. However, the transition tax exemptions provided by the government are time-limited, and if vehicle replacement is not completed by 2030, there will be a risk of increased replacement costs. From 1 January 2025, a 25% discount will be implemented for zero-emission cars, effective until 2029. Consequently, from 2030, zero-emission cars will incur a higher Motor Vehicle Tax compared to gasoline cars, due to their generally greater weight (RVO, n.d.b).

The BFF, as a union, can play a role in gathering the challenges and obstacles faced by members undergoing transformation, providing fact-based data, and engaging in negotiations with local authorities. The film industry's transportation emissions are lower than those of other sectors, such as retail and logistics, so it should not be treated the same way (Interviewee D).

If research shows that the pollution caused by transportation in the film industry is minimal, then the BFF can use this to apply for exemptions or gradually transition, like only replacing diesel vehicles with EVs when their lifespan is reached. One BFF member thinks that for ZEZ regulations, emissions should be based on distance travelled rather than a uniform limit. Highly polluting vehicles can still enter ZEZ, but once they need to be replaced, they must be replaced with EVs. This means that companies that travel more frequently need to make the switch earlier, while less frequent travellers can make the switch later. In this way, the transition is more gradual, and grid congestion can be addressed by further spreading out the transition. In this way, the BFF could leverage its collective strength to enhance communication with municipal authorities, strive for a stronger voice, and participate in discussions on rulemaking beyond the 2030 transition period (Interviewee A). In addition, a more experimental idea is to select a company within the BFF to lead the change, with other members' help and support, to collect insights from the transition process as a foundation for change across the entire industry.

An interviewee also believes that relevant film funds should provide financial support for sustainability initiatives among suppliers. Additionally, advertisers and other upstream clients should increase their budgets for sustainability requirements to facilitate the transition to ZEZ. Finally, to integrate sustainability throughout the filmmaking process, it was suggested to involve eco-consultants and suppliers in the pre-production stage to plan for a greener film in advance, rather than during production, when it is difficult and too fast-paced to change plans (Interviewee C).

4.2E Qualitative Data from Survey

To provide a holistic picture of the situation, qualitative data were also gathered through survey questions, presented in the example survey in Appendix 9.1. The answers reveal the perspective of the BFF members in relation to the transition. Regarding the charging infrastructure requirements to fulfil the needs of the transition to electric vehicles, half the respondents seem to face challenges as they do not own a warehouse, where they could install such technologies, or they are not able to afford renting one. Technical issues were also mentioned in the answers, like power limitations and the net congestion in the municipality of Amsterdam. Only three companies indicated that they are currently able to charge their vehicles without any issues.

Regarding the transition from conventional vehicles to EVs, four companies mentioned the high costs of the investment as a barrier, while having access to charging stations at the warehouses remains an issue. One company noted the lack of suitable vehicle options

currently available on the market. Additionally, two respondents pointed to the limited loading capacity of EVs as a constraint for production needs. Four companies identified the small range of EVs as a major challenge, as this characteristic renders them unsuitable for media production outside the city limits of Amsterdam. This, in combination with the cost, makes them an unattractive option for the BFF suppliers.

One respondent highlighted that the short distances typically travelled within Amsterdam's ZEZ make the transition to EVs difficult to justify, as the high investment yields minimal benefits. This concern was supported by another company. In regard to costs, one member also pointed out the expensive nature of applying for exemptions in different cities, in the case when they are granted separately. Lastly, an important burden is placed on companies located either in or within the limits of the ZEZ, because all their vehicles need to comply with the new regulations. This becomes particularly problematic when vehicles are rented out, as they cannot monitor whether those vehicles enter the ZEZ, and their customers may not be aware of the city regulations.

These findings highlight the complex nature of the transition, involving technical, financial, and logistical challenges that companies need to overcome in the foreseeable future. The next chapter offers a more concrete view of the transition and the investment requirements.

5. Sub-Question 3: Scenario Development

5.1 Methodology

To answer this sub-question, a scenario-based projection analysis was conducted. Three scenarios were developed: a Business-as-Usual (BAU) scenario, a Zero-Emission (ZE) scenario, and a Partial Compliance (PC) scenario. Each scenario models the estimated carbon emissions and financial implications, such as investment costs and fuel costs, associated with different transition pathways. No subsidies were taken into account in the calculations since they are granted on case-by-case criteria and cannot be easily nor accurately estimated. Costs for insurance and the installation of charging infrastructure are also excluded from the calculations. All scenarios consider the upcoming years until 2035 to ensure uniform results. Unlike some economic models that begin with start-of-year values but finalise using end-of-year figures, the approach of the report consistently used beginning-of-year values throughout the analysis.

The BAU scenario assumes no significant changes to the current fleet composition or fuel types, meaning that the existing vehicles will be replaced by the same vehicle type in the future. The ZE scenario explores the ideal transition to a fully zero-emission fleet aligned with regulatory targets by 2030. In other words, all vehicles are upgraded to electric ones by that year. The PC scenario presents a more gradual pathway, balancing emission reductions with realistic financial and operational constraints. The latter assumes flexibility from regulatory bodies, giving the BFF members more time to comply with the regulations. In this case, the fleet shifts from conventional to EVs when their reported upgrades are due; therefore, the shift is completed by 2035. In both the ZE and the PC scenarios, the vehicles are upgraded to EVs.

The projections considered only the adoption of conventional or EVs due to the additional barriers to hydrogen-fuelled truck adoption outlined in Chapter 1 of this report. All three scenarios were created for the distances travelled inside the ZEZ and for the total distances travelled, allowing a comparison between the two cases.

All assumptions involved in the projections are explained in Sections 5.1A and 5.1B below. The calculations were performed per vehicle and were only added up in the end to present a complete picture. In all the scenarios, the maximum lifetime of a conventional vehicle was assumed to be 16 years. This value was estimated based on the highest lifetime observed in the vehicles of the survey responses (a vehicle from Locatiewerk) and helped fill the gaps in the time left until upgrading the old vehicles. For EVs, the lifetime calculated was 15 years (Agrawal, 2023). Moreover, when a vehicle is upgraded to an electric one, it was assumed that the EV is of the same weight type described in Chapter 3 (light, medium, heavy) as the old vehicle and that the distances travelled are the same. Since the EV market is not widespread enough yet, it is assumed that only lightweight EVs could be purchased second-hand, for the companies that chose second-hand in their last vehicle purchase.

It is to be noted that one company is planning to retire from supplier activities in 2029, as stated in the qualitative part of their survey response. Hence, its emissions and costs were considered equal to zero from 2030 forward. Since this company's fleet is composed of only two light vehicles, the overall results are expected not to be affected by their absence in the studied period.

While this analysis aims to provide practical guidance, several uncertainties may influence the outcomes. These include the future availability and geographic coverage of charging

infrastructure for EVs, evolving policy incentives, and the risk of import tariffs on such vehicles. As these factors are beyond the scope of the current analysis and not entirely predictable at this stage, they will be addressed as limitations in the final discussion.

5.1A Carbon Emissions Analysis

The first step of the scenario analysis is the projections of the fleet's carbon emissions. For each scenario, this model used Excel to calculate annual emissions outputs for each vehicle. Emissions from scopes 1 and 2 were included for the transport of the equipment. Scope 1 emissions refer to direct emissions from owned or controlled sources (e.g., tailpipe emissions from diesel trucks), while scope 2 emissions cover indirect emissions from the generation of purchased electricity used by a company (e.g., emissions from electricity used to charge EVs).

Scope 1 is relevant for the current situation, where the fleet is not exclusively composed of EVs. However, in the ZE and PC scenarios, the EVs do not directly emit CO₂. Therefore, the emissions generated come from the electricity generation needed to charge the final fleet, which falls under the scope 2 emissions category.

Scope 1 and 2 emissions were calculated using the following two formulas. The scope 1 emissions of conventional vehicles were estimated using Formula 1.

$$(1) CO_{2,CV} = C_{fuel} \times D \times EF \text{ [t CO}_2\text{]}$$

Where:

- C_{fuel} = Conventional vehicle fuel consumption [L/km]
- D = Annual distance travelled by conventional vehicle [km]
- EF = CO₂ equivalent emission factor per unit of energy [t CO₂/L]

The fuel consumption as well as the distances were obtained from the survey responses. The distance referred to either the total distance or the one travelled inside the ZEZ, depending on the projection. The distances have an assumed annual increase of 2%. This value is based on the predicted increase in the production volume of the Dutch film and television sector (5%) (NL Filmfonds, 2025). Yet, a lower value is taken given that the growth of a company is not directly translated to their travels. Formula 2 was used to calculate the annual kilometres travelled.

$$(2) D_i = D_{2025} \times (1 + r_d)^i \text{ [km]}$$

Where:

- i = Year of calculation [year]
- D_i = Annual distance travelled [km] for year i
- D_{2025} = Distance travelled in 2025 [km]
- r_d = Annual increase rate of distance [%]

The emission factor in the Netherlands is assumed to be constant at 3.25 kg CO₂/L based on the 2025 value (Stichting Stimular, 2025). For the carbon emissions of the EVs, only the indirect emissions caused by the electricity generation are relevant (scope 2). Formula 3 was used for such emissions calculations.

$$(3) CO_{2,EV} = C_{el} \times D \times GE \text{ [t CO}_2\text{]}$$

Where:

- C_{el} = Electric vehicle electricity consumption [kWh/km]
- D = Annual distance travelled by electric vehicle [km]
- GE = Grid emission factor [t CO₂/kWh]

The electricity consumption is assumed constant and equal to 0.25 kWh/km for light vehicles, based on the range given by RouteLogic B.V. (n.d.). For medium vehicles, the consumption used is 1.32 kWh/km. This value uses the assumption that they drive 80% of the time inside the city and uses data from Shoman et al. from urban areas and highways (2023). The heavy vehicles were assumed to consume 1.80 kWh/km (Shoman et al., 2023), given that they mostly travel on highways. The grid emission factor in the Netherlands in 2025 has a value of 0.268 kg CO₂/kWh (Stichting Stimular, 2025). It is assumed to decrease by 1.74% per year, calculated from the CO₂ emissions over the last 23 years (CBS, 2024). Formula 4 below shows the calculation for the grid emission factor.

$$(4) GE_i = GE_{2025} \times (1 - r_{GE})^i \text{ [t CO}_2\text{/kWh]}$$

Where:

- i = Year of calculation [year]
- GE_i = Grid emission factor in year i [t CO₂/kWh]
- GE_{2025} = Grid emission factor in 2025 [t CO₂/kWh]
- r_{GE} = Annual decrease rate of grid emission factor [%]

At the end of the analysis, the annual emissions in each scenario were added up to reveal potential discrepancies.

5.1B Economic Analysis

Alongside the carbon emission analysis, a simplified cost model was applied to offer financial insights that support supplier decision-making. Not all the data required in this step was provided by the BFF. Data on the purchase cost was missing for 22.8% of the vehicles. 12.0% of them were assumed to have the same cost, as provided in the qualitative part of the survey by the respective company. The cost of the rest was found using the RDW official website's catalogue price, via a license plate check, as they were all new vehicles (RDW, 2025). When the price was presented as a range depending on the different vehicle options, an average of the minimum and the maximum value was used.

The yearly costs in the analysis include fuel costs (in the case of conventional vehicles), electricity use costs (in the case of electric vehicles), maintenance costs, and investment costs for vehicle purchases. In the year of the upgrade, the residual value of the old vehicle is assumed to be a profit for the company. The total annual costs are calculated based on Formula 5, as seen below. The calculations were performed by vehicle on a yearly basis. At the end of the analysis, the annual values were added up so that the whole fleet could be presented as one.

$$(5) TC = I_{fuel} + I_{el} + M + I_{new} - R_{old} \text{ [€]}$$

Where:

- I_{fuel} = Fuel costs for conventional vehicles [€]
- I_{el} = Electricity costs for electric vehicles [€]
- M = Maintenance costs [€]
- I_{new} = Investment costs for upgrading to a new vehicle [€]
- R_{old} = Residual value of old vehicle [€]

The yearly maintenance costs were obtained from the survey responses. It was assumed that the maintenance costs are constant and remain the same after the vehicle upgrade. The I_{new} and R_{old} are taken into account only in the year of the vehicle upgrade. For the ZE and the PC scenarios, the investment costs associated with the vehicle purchase in the year of the upgrade were different, depending on the weight type of the electric vehicle. The second-hand EVs cost €28,000. This is based on a Dutch Vans website (n.d.). This price initially excludes VAT but has been adjusted to include it. The new light vehicles cost €55,000, based on Rabobank's range from €50,000 to €60,000 (Rabobank, n.d.b). The new medium EVs cost €230,000, and the new heavy ones cost €330,000 (Nijenhuis, 2024, Rabobank, n.d.a). Based on the drop observed in the period 2016-2024 in the Dutch market for electric vehicles, there seems to be an annual decrease rate of 1.2%. (Statista, 2024). In this report, a 5.0% rate was used instead, given the 10.0% assumed depreciation rate of old fleet vehicles explained below in this section. This is seen in Formula 6, and it is relevant only in the case of EVs, either second-hand or newly bought.

$$(6) I_{new,i} = I_{new,2025} \times (1 - r_{I_{new}})^i \text{ [€]}$$

Where:

- i = Year of calculation [year]
- $I_{new,i}$ = Investment costs for upgrading to an electric vehicle in year i [t CO₂/kWh]
- $I_{new,2025}$ = Investment costs for upgrading to an electric vehicle in 2025 [t CO₂/kWh]
- $r_{I_{new}}$ = Annual decrease rate of investment costs for upgrading to an electric vehicle [%]

In the case of the BAU scenario, the investment costs were assumed to be the same as the original purchase price of the vehicles. In each investment, the cost of installing a special interior to the vehicle was taken into account, if the old vehicle included such a feature.

The fuel costs for the conventional vehicles were estimated using Formula 7.

$$(7) I_{fuel} = P_{fuel} \times C_{fuel} \times D \text{ [€]}$$

Where:

- P_{fuel} = Conventional vehicle fuel price [€/L]
- C_{fuel} = Conventional vehicle fuel consumption [L/km]
- D = Annual distance travelled by conventional vehicle [km]

The fuel costs refer to either diesel or diesel/HVO, depending on the vehicle type in the survey response. The price per litre differs, with diesel having a price of 1.604 €/L in 2025 (CBS, 2025b) and diesel/HVO a price of 1.925 €/L in 2025, based on Schilperoort's (2024) observation that HVO is around 20% more expensive than regular diesel. Those values were

assumed to have an increasing rate of 2.7%, given data on diesel (ChemAnalyst, 2024). This increase is presented in Formula 8.

$$(8) C_{fuel,i} = C_{fuel,2025} \times (1 + r_{C_{fuel}})^i \text{ [€/L]}$$

Where:

- i = Year of calculation [year]
- $C_{fuel,i}$ = Conventional vehicle fuel price in year i [€/L]
- $C_{fuel,2025}$ = Conventional vehicle fuel price in 2025 [€/L]
- $r_{C_{fuel}}$ = Annual increase rate of conventional vehicle fuel price [%]

Fuel consumption was constant, based on the per-vehicle responses from the BFF members. As seen in Section 5.1A, the distances travelled in either the ZEZ or the total ones have an annual growth rate of 2% and can be calculated using Formula 2. The electricity use costs for EVs were found using Formula 9.

$$(9) I_{el} = P_{el} \times C_{el} \times D \text{ [€]}$$

Where:

- P_{el} = Electric vehicle electricity price [€/kWh]
- C_{el} = Electric vehicle electricity consumption [kWh/km]
- D = Annual distance travelled by electric vehicle [km]

The price of electricity was found to be 0.50 €/kWh, considering that companies do not own charging stations (ANWB, 2025b). This might be the case for the BFF members, since they mostly do not use electric vehicles at the moment, some do not own a warehouse, and the grid might not have the potential to accommodate for new infrastructure. The assumed increase rate in the calculations was 2.5% (CBS, 2025a), as seen in Formula 10. The fuel consumption was the same as the one explained in Section 5.1A, depending on the weight type of the EV.

$$(10) P_{el,i} = P_{el,2025} \times (1 + r_{P_{el}})^i$$

Where:

- i = Year of calculation [year]
- $P_{el,i}$ = Electric vehicle electricity price in year i [€/L]
- $P_{el,2025}$ = Electric vehicle electricity price in 2025 [€/L]
- $r_{P_{el}}$ = Annual increase rate of conventional vehicle fuel price [%]

Lastly, the residual value of the old vehicles was calculated using Formula 11.

$$(11) R_{old} = I_{old} \times e^{-dr \times t} \text{ [€]}$$

Where:

- I_{old} = Original purchase cost of old vehicle [€]
- dr = Annual depreciation rate of old vehicle [%]
- t = Year of replacement [years]

The original cost of the old vehicle was found in the survey responses. Vehicles' depreciation rate is 10%, as their price decreases over time, given the time value of money and assets (Univé, n.d.).

All the values used in the projection calculations are summarized in Table 11 below.

Table 11: Summary of projection assumption parameters

Parameter	Symbol	Value	Source
Annual increase rate of distances travelled (both for total kilometres and kilometres in ZEZ)	r_d	2%	NL Filmfonds (2025)
CO ₂ equivalent emission factor	EF	3.25 kg CO ₂ /L	Stichting Stimular (2025)
Electricity consumption for light EVs	C_{el}	0.25 kWh/km	RouteLogic B.V. (n.d.)
Electricity consumption for medium EVs	C_{el}	1.32 kWh/km	Shoman et al. (2023)
Electricity consumption for heavy EVs	C_{el}	1.80 kWh/km	Shoman et al. (2023)
Electricity grid emission factor in 2025	GE_{2025}	0.268 kg CO ₂ /kWh	Stichting Stimular (2025)
Annual decrease rate of electricity grid emission factor	r_{GE}	1.74%	CBS (2024)
Second-hand light EV purchase price	I_{new}	€28,000	Dutch Vans (n.d.)
New light EV purchase price	I_{new}	€55,000	Rabobank (n.d.b)
New medium EV purchase price in 2025	I_{new}	€230,000	Nijenhuis (2024); Rabobank (n.d.a)
New heavy EV purchase price in 2025	I_{new}	€330,000	Nijenhuis (2024); Rabobank (n.d.a)
Annual decrease rate of EV purchase price in 2025	$r_{I_{new}}$	5.00%	Statista (2024)
Diesel price in 2025	C_{fuel}	1.604 €/L	CBS (2025b)
Diesel/HVO price in 2025	C_{fuel}	1.925 €/L	CBS (2025b); Schilperoort (2024)
Annual increase rate of diesel and diesel/HVO	$r_{C_{fuel}}$	2.7%	ChemAnalyst (2024)
Electricity price for EV consumption	P_{el}	0.50 €/kWh	ANWB (2025b)
Annual increase rate of electricity price for EV consumption	$r_{P_{el}}$	2.5%	CBS (2025a)
Depreciation rate for existing vehicles	dr	10%	Univé (n.d.)

5.2 Results & Discussion

5.2A Carbon Emissions Analysis

Based on the methodology explained in Section 5.1A, Figures 5 and 6 present the annual carbon emissions throughout the studied period (2025-2035) for the three scenarios: business-as-usual (BAU), zero-emission (ZE), and partial compliance (PC).

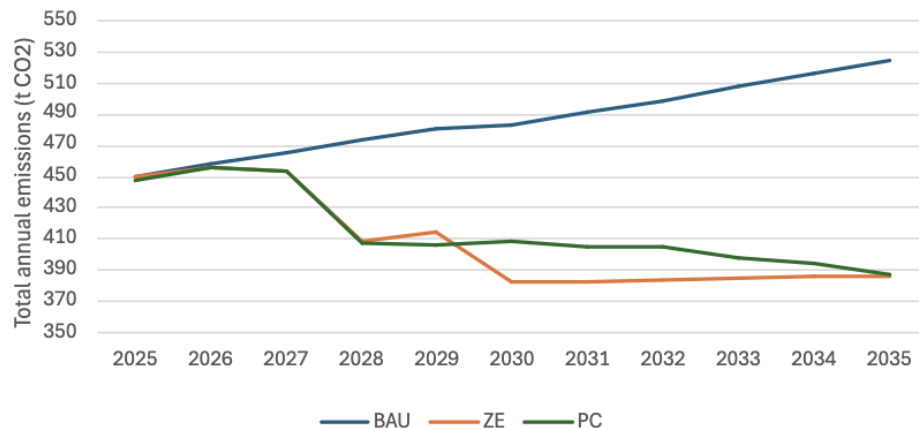


Figure 5: Total carbon emissions from total fleet distances travelled under the three scenarios (t CO₂)

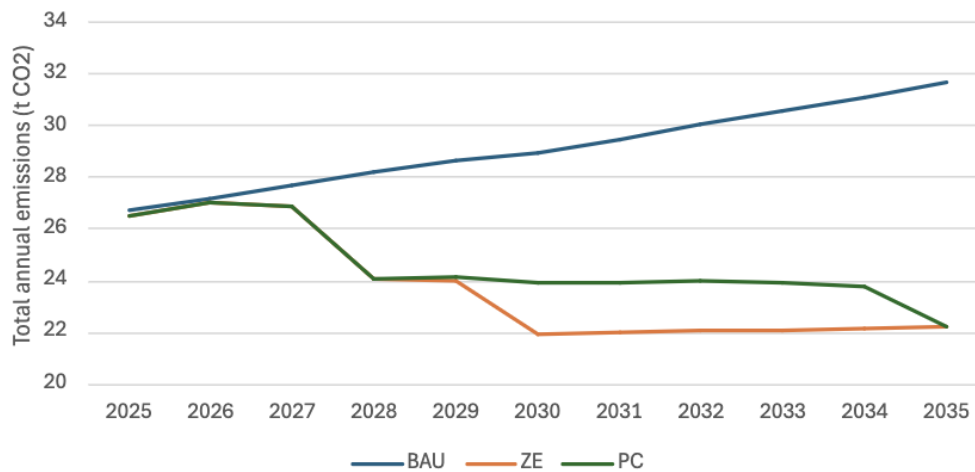


Figure 6: Total carbon emissions from total fleet distances travelled in zero-emission zones under the three scenarios (t CO₂)

As seen in the figures above, the fleet emissions in the ZEZ are only a small segment of the emissions generated in the total travelled distances. As expected, the BAU scenarios present an increase in emissions with time, since the old diesel and diesel/HVO vehicles are replaced with others of the same type. So, no transition is achieved. Only three electric vehicles were considered, which are already part of the current fleet. The increase in emissions could be justified by the assumed rate of increase of 2% in the kilometres travelled by the vehicles.

Both the ZE and PC scenarios depict a transition towards a more sustainable fleet, as desired by the regulations discussed in the previous chapters. In 2028, there seems to be a significant drop in emissions in the ZE and PC, coming from the fact that multiple fleet vehicles are expected to be upgraded in that year, as seen from the survey responses.

The ZE and PC scenarios end in the same emission values, since in both cases the fleet is composed exclusively of electric vehicles in 2035. In the ZE scenario, an abrupt reduction of emission is also observed in the period 2029-2030, as the fleet is expected to have upgraded

every vehicle to an EV by that time. Even though this phenomenon is desirable, its economic feasibility is to be questioned and is discussed in Section 5.2B below. From 2030 until 2035, the slight rise of emissions in the ZE scenario stems from the assumed increase rate of 1.74% of the grid emission factor in the Netherlands. The PC scenario offers a more gradual emission reduction, presenting a small annual decrease until the end year.

5.2B Economic Analysis

Figure 7 below was created based on the methodology explained in Section 5.1B above. It shows the total costs and total final energy consumption (FEC) costs, including fuel and electricity, under the three scenarios: BAU, ZE, and PC. Each bar represents the total cost, with the darker segment indicating the portion attributed specifically to FEC costs.

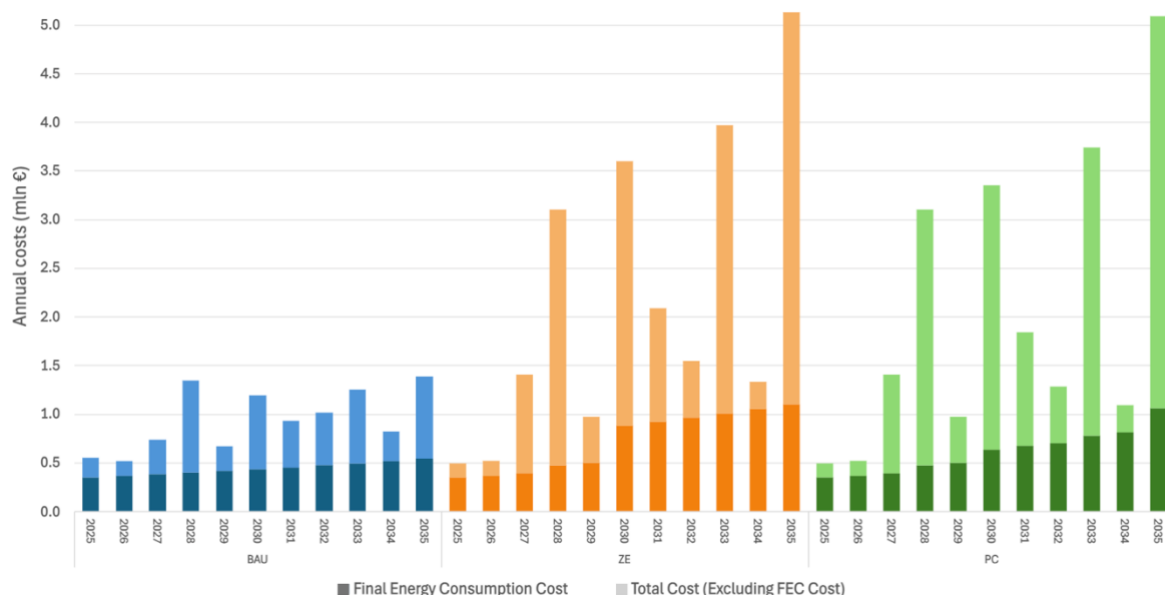


Figure 7: Total costs and total final energy consumption costs (fuel/electricity) under the three scenarios (mln €)

It is observed that the BAU scenario involves the lowest costs. This could be justified by the lower investment cost of conventional vehicles compared to the high costs of EVs. The not-yet-established market of second-hand EVs aggravates the situation. As explained in the methodology, it is presumably possible to have access to second-hand EVs only for the light-weight vehicle category.

The PC scenario presents similar or lower annual cost values compared to the ZE scenario. The investment costs of EVs were considered to be decreasing in the future, giving the PC scenario an advantage. The research period also plays an important role in the financial aspect of the transition. In this case, only a 5-year period after the 2030 'deadline' imposed by the regulations was given, with a great degree of uncertainty. If this period were further extended, the results would differ, giving the BFF members more time to invest in the fleet upgrade.

Moreover, total investment costs for ZE or PC scenarios could decrease significantly if heavy trucks were allowed to remain conventional, provided they meet a very high emission

standard, since they rarely enter ZEZ. This is particularly relevant given the currently high costs of electric heavy-duty trucks.

In the BAU scenario, where the fleet remains mostly composed of conventional vehicles, as well as in the PC scenario, where the conventional vehicles are used for a longer period of time, the FEC costs are lower than the ZE scenario. This is due to the limited fuel consumption by certain conventional vehicles of the fleet, in combination with an elevated electricity consumption by the medium and heavy vehicles. For such financial factors, the assumptions used influence the outcomes a lot, even though the investment costs have a greater impact than the final energy consumption.

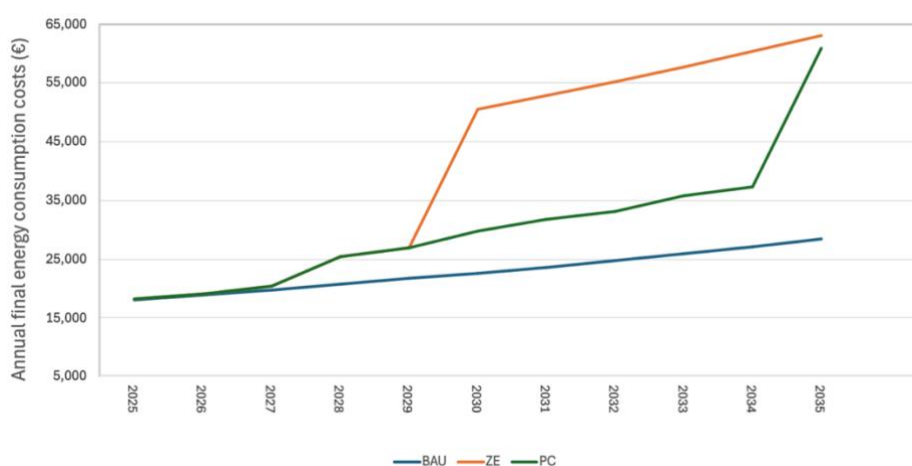


Figure 8: Annual final energy consumption cost from distances within zero-emission zones under the three scenarios (in €)

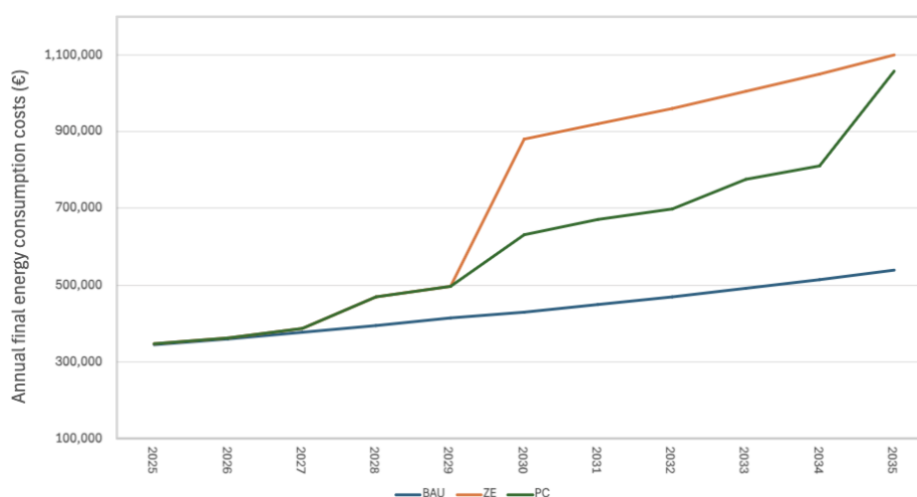


Figure 9: Annual final energy consumption cost based on total fleet distances travelled under the three scenarios (in €)

From Figures 8 and 9 above, it can be observed that, under the BAU scenario, the overall cost of FEC does not rise sharply, but rather shows a moderate growing trend over time. This indicates that continuing with conventional vehicles may not lead to significant cost increases in a short-term period, but it is hard to predict the diesel price in the long-term market trend. As for the other two scenarios, the ZE scenario grows faster because of the demand to

complete the transition of the whole fleet to EVs by 2030. So, when the fuel is changed from diesel to electricity, the amount of final energy consumed for the same unit of distance travelled increases. Therefore, although the unit price of electricity is approximately 1/3 of the price of diesel per unit, the cost of electricity will become higher and higher as the distance travelled increases.

For the PC scenario, the fuel cost grows more gradually from 2029 to 2034 with a certain time buffer, but it experiences a sharper rise between 2034 and 2035. This could be because most companies have a longer upgrade cycle, typically around 10 years, which will lead to a concentrated replacement of vehicles with EVs during this period, thus accelerating the rapid shift in fuel type and electricity costs around 2035.

6. Recommendations

Although ZEZ were announced several years ago to give suppliers time to adapt, the pace of development in the EV market and supporting infrastructure has not kept up, making the transition to zero-emission compliance particularly challenging. Supplier companies in the media industry operate vehicles with particular requirements, such as heavy-duty special interiors, which are not yet fully supported by available EV options. Switching between diesel and electric vehicles depending on whether operations occur inside or outside ZEZ, or across different filming locations, sometimes outside the Netherlands, adds complexity.

While exemptions to the regulations exist, applying to them is complicated for small companies like BFF supplier companies, which lack the capacity or time to navigate regulatory channels. Moreover, the application process must be handled on a vehicle-by-vehicle basis, further complicating an already complex process.

Therefore, to ensure a fair and effective transition to ZEZ compliance the following recommendations are proposed:

- **Provide targeted support** from regulatory agencies to sectors with specialised vehicle requirements. This support could include the creation of a centralised and simplified exemption process for small companies, helping level the playing field with larger firms that have greater administrative capacity.
- **Extend transition timelines** for companies facing technical limitations, particularly those using vehicles that already meet high emission standards and operate only partially within ZEZ, would help make the transition more equitable for small operators affected by current market and policy gaps.
- **Adopt a mixed-fleet strategy** to optimise costs and compliance. The companies that have a large fleet could partially use electric vehicles, when traveling within ZEZ and continue using conventional vehicles outside of these zones. This approach allows for a more gradual transition with less upfront investments.
- Related to this, companies can decide to **delay part of their investments** until EV prices drop further. Battery and electric vehicles prices are decreasing steadily. By waiting with some purchases, companies have a wider market available, with improved models with extended ranges. By this time the second-hand market should be expanded, which could then also be considered as a viable alternative.
- **Plan charging infrastructure** well in advance. Companies should anticipate long wait times for charging infrastructure and possible grid congestion in these areas. Early planning for the number, distribution and power supply of charging stations is very important.
- **Improve understanding of policy information** amongst suppliers. Information such as subsidies for EVs, charging infrastructure and zero-emission policies etc., which may help them arrange transition strategies with more possible cost-effective chances. If conditions permit, BFF members could also seek the expertise of consultants to provide professional interpretation and strategic advice.
- **Co-invest in charging infrastructure.** Companies whose locations are close in proximity could invest in charging infrastructure together. This collaboration reduces the cost of the investment. This is especially beneficial for companies with a medium-sized fleet. By getting their own charging stations, they also have a reduced cost of charging, since for privately owned charging stations, they can pay their normal electricity price (± 0.27 €/kWh), instead of the prices for public charging stations (± 0.50 €/kWh) (ANWB, 2025a; ANWB, 2025b).

- **Collaborate with the energy sector** to help solve the local grid congestion issues. Early on, collaboration can help ensure that charging infrastructure plans are aligned with the current and future grid capacity. This reduces the risk of delays.
- **Make collective negotiation strategies.** BFF, either independently or together with other companies, could negotiate with the municipality of Amsterdam to get the exemptions needed, collectively or get extended timelines based on their relatively low share of emissions within ZEZs.

These recommendations can help BFF members and other stakeholders navigate the road to zero-emission compliance in a way that is feasible and financially sustainable.

7. Limitations and Future Research

7.1 Quantitative Data Limitations

The data obtained from BFF represents most of the fleet since 9 out of the 18 companies were involved in the research. However, it remains unclear whether incorporating the others would give significantly different outcomes. Therefore, the findings can only be considered reliable for this subset of the total fleet.

The quantitative data gathering process itself during SQ1 involves certain risks. Its reliability is dependent on the structure of the survey, the participants' interpretation, and the accuracy of the responses (Queirós et al., 2017). Even though the participants were further contacted in case the data did not seem accurate, this cannot guarantee complete data reliability. In particular, some questions required calculations from the part of the BFF members that cannot be verified by the research team (e.g. the distances travelled in ZEZ).

The inevitable presence of missing data in the responses is another limitation of the study. Since the members were free to not fill in questions, missing information was present in the final responses. To answer the gaps in the data, assumptions were made based on the rest of the fleet or using online sources. Furthermore, this study did not apply annualization to the investment cost of EVs, so it failed to consider the time value of money over the lifetime of EVs and the effect of interest (inflation). Thus, further research is advisable to add more advanced financial evaluations such as Net Present Value and Internal Rate of Return.

Moreover, certain factors in the future projections involve an inherent level of uncertainty that cannot be easily estimated. For example, the projections of fuel and electricity prices in the model were based on current price assumptions or fixed growth rates, so the potential impact of future market fluctuations (especially the electricity market) has not been taken into account. Likewise, the waiting times for the installation of charging infrastructure for EVs are unknown and are case-dependent. Lastly, it is to be considered that the BFF is a union, yet each company can make independent decisions regarding its future strategies. To illustrate that, one company could invest in EVs right away to avoid the administrative burden of applying for exemptions, and another company could try to get the exemptions, in case vehicle upgrades are out of its budget.

7.2 Qualitative Data Limitations

The qualitative part of the research was primarily based on the interviews outlined in Chapter 4, and only a small part of the survey. While both contain risks of misinterpretation between the respondent and the receiver, the interviews, in particular, present inherent limitations to the study. Even though they are a time-consuming process, they are able to represent only a small portion of the stakeholders (Alshenqeeti, 2014). Therefore, the limited number of interviews conducted can never show a complete image of reality since not all voices are heard, even though they can offer great insight into the situation.

Subconscious biases of the interviewers can also alter the outcomes of the interviews (Adams-Quackenbush et al., 2019; Alshenqeeti, 2014). The team tried to remain neutral. Nevertheless, the whole team was aware of the situation that the BFF is facing as well as their hurdles, creating a possibility that the mindset during the formulation of interview questions and the interview process itself may have leaned toward a more negative perspective. While answering SQ2, the use of open-ended questions allowed participants to give longer responses, take an active role in the interview process, and offer their holistic perception (Powell et al., 2012), hence, potentially reducing such biases. Yet, biases from the participants should also be taken into account, as they could respond in a way they think the interviewer expects, following the social desirability principle (Larson, 2019).

Moreover, although online sources were consulted to verify statements made during the interviews, some inaccuracies may remain, as not all details can be fully captured or confirmed. This illustrates the broader methodological risk in qualitative research, where findings are shaped not only by what is said but also by how it is interpreted, filtered, and contextualised throughout the research process.

7.3 Future Research

As of now, the BFF have been researched as one fleet. Creating scenarios and making projections for individual companies would also be beneficial to gain more detailed insights into how each should approach the situation. This approach would allow for the inclusion of financial potential, technical considerations, and company-specific circumstances facilitated by the possibility of close collaboration with each company. A company-specific analysis would also improve data reliability as it would be easier to track the steps before the provision of the final values used in the projection calculations. For example, by using GPS tracking systems in all trucks of a company could give accurate estimates of the distances travelled in the ZEZ of Amsterdam. In any case, the inclusion of discount rates would help the annualization of the expenses/benefits and allow for a more accurate depiction of reality, accounting for the time value of money.

Conducting more interviews could also help gain more insights and create a more realistic picture of the current and potential future situation. Involving more members of BFF in the interview process would also allow for more information on the obstacles they are facing, which was not done in this study due to time constraints.

Replicating this study using data from other industries is feasible and valuable. It could give insights into how implementing the regulations around ZEZ influences a wider segment of the Dutch economy. Since the media industry is not the only sector undergoing this significant shift, examining others is equally important to gain a comprehensive understanding.

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9. Appendix

9.1 Survey Example

The survey was custom-made for each BFF member, as they had already sent some of the information we asked for to another task force member. The survey was developed in a way that made it easier for them to see what they had already sent out and what was missing for this research. Each cell that needs to be filled is grey, implying there is data missing. Once the cell is filled, it turns green to make it easier for them to see whether they need to give more information. Figure 10 below shows an example of how the survey looked like for one company.

Wagenpark Data																	
Nr.	Gekocht of Geleased	Kenteken	Bouwjaar	Euro standaard	Brandstof type	Brandstof gebruik (l/km)	Aanschafkosten (€)	Is het voertuig tweedehands?	Jaarlijkse onderhoudskosten (€)	Aantal jaren tot vervanging	Bijzonderheden aan het voertuig	Kosten bijzonderheden (€)	Gemiddelde maandelijkse reisafstand (km)	Aantal maandelijkse kilometers in zero-emissie zones (km)	MTH (kg)	Lengte (m)	Opmerkingen:
1		00-BTF-S	2022	6	Diesel/HVO						Ja				7500	7,35	Vrachtwagen, Bakwagen, Magazijn inrichting
2																	
3																	
4																	

Extra data	
1	Zou u de telematica gegevens van uw voertuig willen delen? Zo ja, zou u de link hiernaast kunnen plaatsen:
2	Als u een nieuw voertuig aanschaft is dit dan gewoonlijk een tweedehands of een nieuw voertuig?
3	Wat is uw gemiddelde investeringsbudget voor een nieuw voertuig?
4	Heeft u de mogelijkheid elektrische voertuigen op te laden op locatie/kantoor? Indien niet wat is hier toe de reden?
5	Heeft u elektrische voertuigen overwogen voor uw wagenpark? Indien niet, wat zijn uw grootste bezwaren/belemmeringen tot aanschaf van een elektrisch voertuig?
6	Wanneer u kijkt naar de zero-emissie zones in Amsterdam, wat zijn voor uw organisatie de grootste uitdagingen?

Commerciële data	
1	Kunt u een inschatting maken van de gemiddelde hoeveelheid kilometers die uw voertuigen afleggen voor onderstaande activiteiten?
	Reclames
	TV
	Film
	Anders:
2	Hoeveel procent van uw werk valt ongeveer in de volgende categorieën?
	Reclames
	TV
	Film
	Anders:

Figure 10: Survey example

9.2 Taskforce Interview Questions by Stakeholder

Film Producer

- 1) Could you explain a bit about the organisation you work for and how you are affected by the implementation of the zero-emission zones?
- 2) What challenges or barriers do you foresee in adapting to the zero-emission regulations by 2030?
- 3) Do you feel the film production sector has had enough opportunity to provide input on these regulations?
- 4) What kinds of support or policy adjustments would most help the film sector meet these sustainability goals?
- 5) In what way forming an alliance, like NAPA, can help the film industry comply with new policies or even ask for regulation adjustments?

Officer from the Ministry of Infrastructure and Water Management

- 1) Could you shortly explain about your role within the ministry?
- 2) What role does the ministry play in guiding/regulating municipalities in implementing zero-emission zones? Do most of the regulations come from the municipality itself or from the national government?

- 3) Are there current or upcoming funding instruments or incentive programs to help sectors transition to zero-emission transport? Or is there a way the ministry can help sectors transition to meet these new regulations?
- 4) What exemptions have been considered so far, and what are the criteria?
- 5) Is there any consideration for exemptions for the film industry? When will that be certain?
- 6) Are you aware of any feasibility studies conducted before implementing the zero-emission zones in Amsterdam?
- 7) Are you in contact with the electricity suppliers to plan how the net congestion will look like by 2030?

Officer from the municipality of Amsterdam

- 1) From your perspective, what logistical or operational challenges are most common for companies that want to comply with the zero-emission zones?
- 2) What main barriers do you see for the members of the BFF with complying to the zero-emission regulations?
- 3) Is the municipality working together with the ministry (of infrastructure and water) for policy implementation?
- 4) What exemptions have been considered so far, and what are the criteria?
- 5) What has been the approach so far for granting exemptions or flexibility during the transition period?

Member 1 from BFF

- 1) What main barriers do you see for the members of the BFF with complying to the zero-emission regulations?
- 2) What would be the main barriers and challenges when supplier companies do want to renew their fleet?
- 3) How has BFF engaged with the municipality or other stakeholders to address these issues?
- 4) Do you feel that other sectors are receiving preferential treatment or support compared to the film supply industry?
- 5) What would an ideal solution or exemption look like for your sector?
- 6) Does the formation of BFF make matters easier when it comes to complying with policies?
- 7) Are BFF members' aspirations and challenges aligned? Will others face more issues when it comes to complying with the regulations?

Member 2 from BFF

- 1) Can you describe how your company typically operates when entering city centres like Amsterdam?
- 2) What are the main challenges of complying with the zero-emission regulations?
- 3) Have you considered fleet renewal or other solutions? and what barriers do you face in doing so?
- 4) What would be the impact on your business if no exemption or support is offered?
- 5) Do you feel your needs are understood or considered by policymakers?
- 6) Was there enough time to comply with the regulation when they were first announced? Is there enough time now?

Member from The Netherlands Film Fund

- 1) Could you explain a bit about the organisation you work for and how you are affected by the implementation of the zero-emission zones?
- 2) Is there a role for the Film Fund in helping the sector transition, such as through targeted grants or sustainability bonuses?

- 3) Could you describe a potential future of the Dutch film industry with the new zero-emissions policies in mind? Are the outcomes mostly positive or are there undesired consequences?
- 4) Has your organisation been affected before by a policy implementation? How was the situation handled?

9.3 Cynefin Table

Domain	Domain sub-themes	Situations
Simple	Ordered, repeatable. Clear cause and effect. Evident to everyone that there is a right answer. Decision is to: Sense the situation – Categorise it – Respond according to best practice. Businesses follow standard operating procedures. Or undergo process re-engineering to meet standard operating procedures.	When the situation faced is mainly covered by the national and local exemptions. The transition phase is sufficient to coordinate with national and/or local authorities to make the switch possible and feasible. Exemption, subsidies, and logistic brokers procedures and costs.
Complicated	Ordered, discoverable. Cause and effect are not evident to everyone, hence there is more than one right answer. Decision model is to: Sense situation – Analyse it – Respond according to good practice. Businesses can undergo scenario planning or integrate systems thinking in their processes.	Case-specific issues that require co-ordinating with a third party (logistic brokers, municipalities) to for exemptions, subsidies, or advice in a smoother electrical transition. Applying for grid connection. Building charging infrastructures and costs. Applying for an exemption is expensive, time-consuming, and difficult to implement (especially in bulk, and case specific scenarios). Re-organisation of business operations due to the switch to electric vehicles.

Complex	Unpredictable. Cause and effect seen in retrospection. Experimentation with what the right solution is. Decision model is to: Probe the situation – Sense the outcomes – Respond accordingly. Stakeholders can collaborate to get a better understanding of the complexity of the situation and have a system-thinking perspective	Cultural shift towards concerns over costs and politics. Further investment into research and development of electric vehicles. Lock-in with transition options of the BFF members. Cooperation with different stakeholders (grid operators and municipalities) for case-specific scenarios not covered by the exemptions.
Chaotic	Turbulent No clear cause and effect. There is no right answer or solution. Decision model is to: Act on the situation at hand – Sense the results and situation – Respond accordingly. Stakeholders and businesses implement crisis management.	Cultural shift towards concerns over costs and politics. Further investment into research and development of electric vehicles. Lock-in with transition options of the BFF members, makes the situation unpredictable and difficult to manage.